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Professional Summary

Senior Machine Learning Engineer with experience in developing and deploying **ML, Generative AI and custom vision solutions**. Expertise in **LLMs, Multimodal RAG, Predictive modeling, MLOps** and **cloud platforms** like **GCP**. Proven success in building AI-powered applications for healthcare, e-commerce and manufacturing industry with innovative problem-solving and leadership skills.

Experience

Senior Machine Learning Engineer | Quantiphi Analytics Pvt. Ltd., Bangalore

Aug 2022 – Present

- Developed and deployed **ML models** for predictive analytics and Generative AI.
- Integrated **LLMs** for advanced solutions in healthcare and e-commerce domains.
- **Led teams of data scientists**, improving project delivery and client satisfaction.
- Developed **production-ready CICD MLOps pipelines** on **GCP** to boost operational efficiency for healthcare providers.
- Designed and implemented **custom ML solutions** for various industries.
- Built predictive analytics models, data preprocessing pipelines, and deployed solutions at scale.
- Delivered **Generative AI** projects, including LLM-driven applications.
- Worked on **sensor data** of manufacturing industry
- Built an advanced **Computer vision** model to detect defects in the chipset manufacturing.
- Built **agentic AI** systems with secure tool-calling and guardrails for production.
- **Productionized LLMs** with retrieval, evaluation, and self-correction workflows.
- Integrated agents with observability/incident tools for auto-diagnostics and remediation.
- Implemented validator models, policy checks, fallbacks, and circuit breakers for reliability.
- Designed RAG pipelines with data validation, feedback loops, and performance optimization.

Technical Skills

- **Languages:** Python, SQL, Bash
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Generative AI, LLMs, Prompt Engineering, RAG, Transformers, Vision-Language Models, Object Detection, Image Segmentation, Deep Learning (CNN, RNN, U-Net, U-Net++), XGBoost, AutoML, SHAP, A/B Testing, Time-Series Forecasting (ARIMA, LSTM)
- **NLP & LLM Ecosystem:** Gemini 1.5 Pro, OpenAI GPT, LangChain, Hugging Face Transformers, Vector DBs (FAISS, Pinecone), Token Optimization, Few-shot/Zero-shot Prompting, PII/PHI Masking (Google DLP), Text-to-Image Embeddings, Retrieval-Augmented Generation (RAG), Gemini Vision
- **MLOps & Model Deployment:** Vertex AI Pipelines, Kubeflow, TensorFlow Extended (TFX), ML Metadata Tracking, Vertex AI Model Monitoring, TensorFlow Serving, Docker, CI/CD (Cloud Build, GitHub Actions), Feature Store, Model Registry
- **Cloud Platforms:** Google Cloud Platform (Vertex AI, BigQuery, Cloud Functions, Cloud Scheduler, AutoML, Looker, GCS), AWS (S3, EC2 – basic exposure)
- **Data Engineering & Visualization:** Pandas, NumPy, Apache Beam (basics), Spark (basic exposure), Data Validation, Looker, BigQuery, Matplotlib, Seaborn, SQL Optimization
- **Tools & Frameworks:** TensorFlow, Keras, Scikit-learn, OpenCV, PyTorch (basic), Git, LabelMe, Streamlit, FastAPI, Docker, Airflow (basics)
- **Dev & Experiment Tracking:** Git, DVC, Weights & Biases (W&B), MLflow (explored)

Key Projects

Chipset Defect Detection using Image Segmentation (U-Net & U-Net++)

- Engineered segmentation masks using custom class mapping and label encoding for multi-class defect detection.
- Trained and fine-tuned U-Net and U-Net++ architectures using Keras and TensorFlow, achieving precise localization of micro-defects. Enabled automation of visual QA, reducing manual inspection effort and enhancing defect traceability.
- Applied advanced data augmentation and optimized IoU and Dice loss metrics for accurate segmentation of small defects.

Multimodal, E-commerce product recommendations(LLM + RAG)

- Used Gemini 1.5 Pro for image-to-text generation and contextual product description creation.
- Engineered multimodal embeddings combining CLIP-based visual and BERT-based text vectors for personalized retrieval.
- Implemented Retrieval-Augmented Generation (RAG) pipeline for product recommendations based on user queries.
- Deployed solution on GCP Vertex AI and integrated with e-commerce platform APIs for real-time product suggestions.

Sensor Data Analysis for Downtime Reduction, (Manufacturing Data Engineering)

- Analyzed 1.2M+ time-series sensor logs to identify downtime patterns in manufacturing equipment (rolling mill, furnace).
- Selected 96 out of 1600 tags using SHAP-based feature selection and correlation heatmaps.
- Implemented alert-based logic and Looker dashboards powered by BigQuery to notify potential machine failures proactively.

Medical Entity Extraction & Clinical Prediction, MLOps automation for healthcare, predictive models, Gen AI engine (Gemini)

- Built an LLM-based data pipeline using Gemini Pro models for extracting clinical entities from raw physician notes.
- Masked PHI/PII using Google DLP and formatted structured outputs into HL7-compatible schema for EHR ingestion.
- Designed CI/CD ML workflows using Vertex AI Pipelines, Kubeflow components, and Cloud Scheduler.
- Used TFX and Data Validation for preprocessing, schema checks, and transformation consistency across pipeline stages.
- Enabled container-based deployment with Cloud Build and Artifact Registry, reducing model iteration time by 40%.
- Implemented drift detection and latency monitoring using Vertex AI Model Monitoring and custom alerting hooks

Agentic AI Framework and MCP Integration

- Integrated Enterprise AI: Developed JIRA/Confluence connectors to ingest and process enterprise data, enabling Retrieval-Augmented Generation (RAG) for AI-driven insights.
- Pioneered Generative AI Features: Implemented advanced image generation and editing capabilities using Google Gemini within the platform.
- Automated Workflows with AI: Created an AI agent server leveraging LLM Function Calling to enable natural language control of complex JIRA operations, significantly boosting productivity.

AKI Risk Prediction using XGBoost, AutoML, and BQML

- Benchmarked 10 ML models on GCP for Acute Kidney Injury (AKI) prediction using clinical records.
- Trained and evaluated XGBoost, AutoML Tables, and BigQuery ML with explainability using SHAP and AUC/F1 as metrics.
- Deployed models via Vertex AI Pipelines with a focus on reproducibility and monitoring performance post-deployment.

Video stitching and object detection for broadcast channel through Custom YOLO V8 model and Open CV SIFT

- Performed real time video stitching to produce a streamable broadcast ready video from 7 cameras placed in different angles and applied object detection to detect player and ball movement.

Key Impact Metrics in Experience

- Reduced human manual defect detection time by ~98% and achieved 93%+ accuracy on defect detection by optimizing the U-Net model for micro-defect segmentation using custom augmentations.
- Reduced model training and deployment cycle time by 40% via automated CI/CD pipelines using Vertex AI, TFX, and Cloud Build.
- Cut manual medical note review time by 80% through structured output generation using LLM-based entity extraction pipeline.
- Achieved 91%+ AUC in AKI risk prediction benchmark, outperforming baseline models and enabling faster triage decisions.
- Minimized unplanned machine downtime by 15–20% by identifying key sensor thresholds and visualizing alerts on Looker.

Education

B.Tech (Hons.)| Lakshmi Narain College of Technology, Bhopal
2018 – 2022

Hobbies

Market Analysis (NSE, Cryptocurrencies) | Traveling | Reading

Certifications

- 1) Google certified [Generative AI Leader](#)

3) Google Certified [Professional ML Engineer \(GCP\)](#)

5) Google Certified [Associate Cloud Engineer](#)

7) **SPOT Award, Quantiphi**
- 2) 17 GenAI Badges by **Google**

4) [L400 - Build and deploy Generative AI solution using RAG](#)

6) **Unsung Hero Award, Quantiphi** — Recognized for consistently delivering high-impact ML and Gen AI solutions

Leadership & Mentorship

- Led a team of 3–4 data scientists across multiple ML and GenAI projects, guiding sprint planning, code reviews, and deployment practices.
- Mentored new joiners and interns on topics like LLMOps, image segmentation workflows, and GCP-based model deployment.
- Took ownership of end-to-end solution delivery, serving as the client SPOC for high-priority healthcare and manufacturing engagements.

